Deluge
MDD Snake + Plaits

An alternative sequencer and additional synthesis engines for OLED Deluge

This custom firmware adds two creative tools to Deluge: **MDD Snake**, a 4x4 sequencer with independent note and modulation lanes, and **Plaits**, an additional oscillator source offering three synthesis engines.

Keep using Deluge's normal sequencer, or switch an individual clip to MDD to explore sequences with different lengths, speeds and movement patterns.

Release	Details
Base firmware	Deluge Community Firmware 1.3.0 beta - Derbyshire
Interface	OLED Deluge
Status	Unofficial custom firmware; not an official Synthstrom release
Validation	ARM build and software tests passed. Comprehensive physical-device testing has not been completed.

What is inside

MDD Snake: seven independent lanes, grid editing, direction and shape controls, randomization, Custom mapping and WRITE with Undo/Redo.

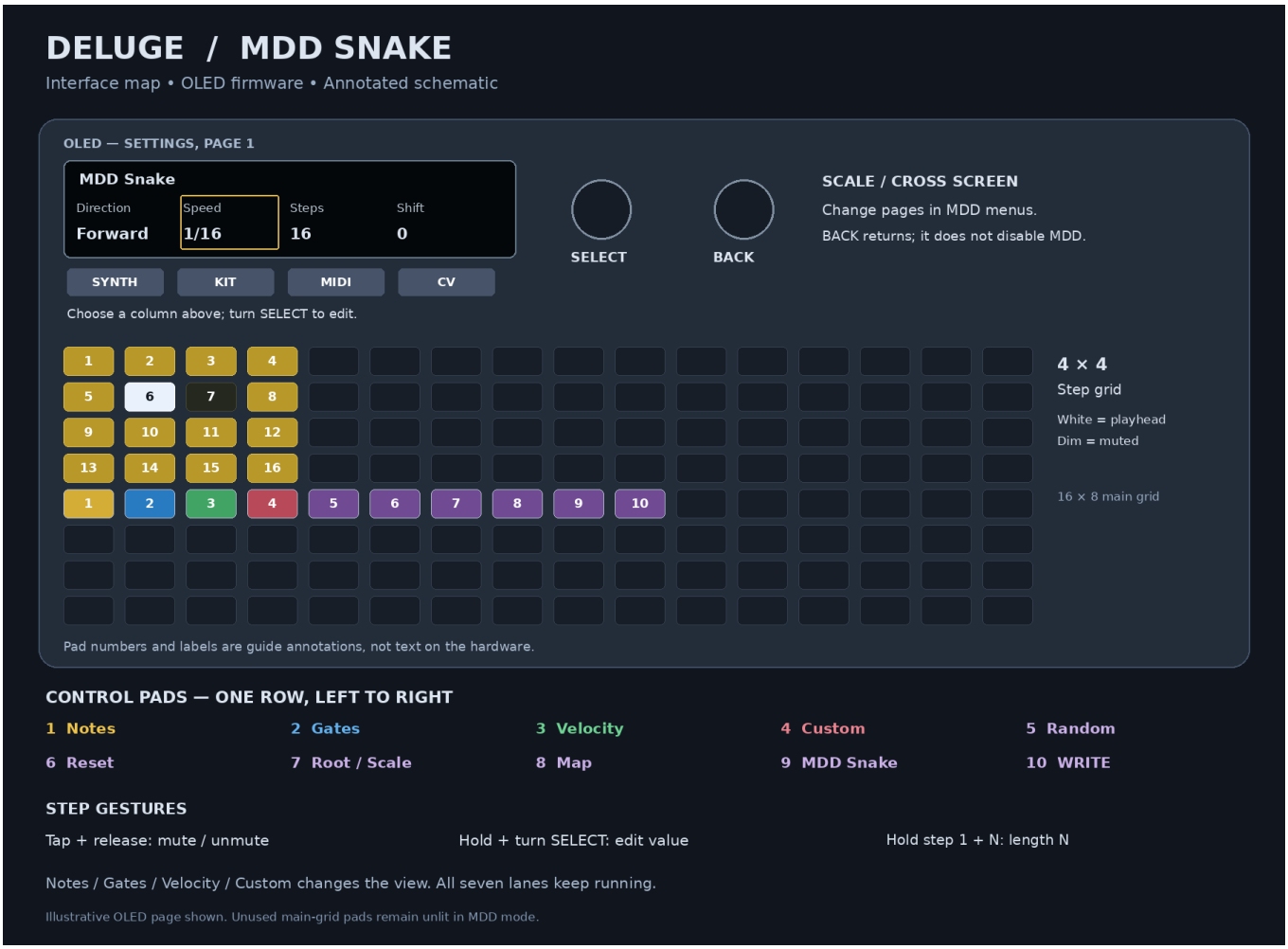
Plaits: FM, Waveshaping and Virtual Analog engines, added alongside the existing oscillator sources.

Additional changes: selected OLED improvements, startup greeting and delay, granular and digital reverb fixes.

Reading guide: Interface map (2) / Sequencer overview (3) / Controls (4) / Rhythm and mapping (5) / Plaits and WRITE (6) / Release notes and limits (7) / Bug fixes (8) / Plaits integration (9).

MDD interface map

The active step grid occupies the upper-left 4x4 area of the main pad grid. Ten control pads sit directly beneath it, in one row from left to right.



Annotated schematic, not a hardware photograph. Numbers and control names are guide labels, not text printed on the pads. The OLED shows an illustrative Settings page; the drawing is not to hardware scale.

Visual state	Meaning
Coloured cells	Steps in the selected lane
White cell	Current playback step
Dim cell	Muted step
Unlit main-grid pads	Unused by the MDD interface

Selecting Notes, Gates, Velocity or Custom changes the lane you view and edit. **All seven lanes continue running.**

One grid, seven independent lanes

Each lane contains up to 16 steps and can have its own **length, speed, direction, shape and shift**. A 16-step melody can run alongside a five-step velocity pattern and a seven-step filter modulation pattern.

Lane	Function
Notes	Notes and note triggers
Gates	Note durations
Velocity	Note intensity
Custom 1-4	User-assigned synth modulation

Movement options include **Forward, Reverse, Pendulum and Random**. Grid paths include **Rows, Columns, Snake Rows, Snake Columns and Spiral**. The current cell is highlighted; muted steps are dimmer.

Quick start

1. Open a synth clip and select **Sequencer > MDD** from the OLED menu.
2. Press the **Notes** control pad.
3. Hold a step and turn **SELECT** to change its note.
4. Select **Gates** and **Velocity** to edit durations and accents.
5. Open **MDD Snake** to set direction, speed, length and grid shape.
6. For modulation, select **Custom 1-4**, edit its steps, then open **Map** and choose a target and Depth.

On the first activation of a new or loaded clip, all seven lanes start at **1/16**.

Mode switching

Normal and MDD sequence data are stored separately. **Sequencer > Normal** returns to the previous Normal sequence; **Sequencer > MDD** returns to the MDD lanes. BACK closes menus without disabling MDD. WRITE deliberately replaces the Normal notes - see page 6.

Controls at a glance

Pad	Control	Purpose
1	Notes	Edit the note lane
2	Gates	Edit note durations
3	Velocity	Edit note intensity
4	Custom	Select and edit Custom 1-4
5	Random	Open Random / Random All
6	Reset	Open Reset / Reset All
7	Root / Scale	Edit Song root and scale
8	Map	Assign Custom targets and Depth
9	MDD Snake	Open sequencer settings
10	WRITE	Transfer notes to the Normal grid

Random and Reset pads open their menus. Select the relevant action and press SELECT to execute it. **All** applies the operation across the lanes.

Step action	Gesture
Mute / unmute	Tap and release
Edit value	Hold the step and turn SELECT
Open value window	Hold for at least 350 ms
Set lane length	Hold step 1, then press the desired final step

After turning SELECT to edit a held step, releasing it does not also toggle mute.

OLED navigation: SYNTH / KIT / MIDI / CV selects one of four parameter columns. Turn SELECT to edit. SCALE / CROSS SCREEN changes pages. BACK closes the window or returns to the previous level. These column controls apply inside MDD menus.

Rhythm, scale and mapping

Window	SYNTH	KIT	MIDI	CV
Notes / Gates / Velocity	Step	Step Value	Steps	Shift
Custom	Custom 1-4	Step	Step Value	Steps
Settings: page 1	Direction	Speed	Steps	Shift
Settings: page 2	Snake Shape	Sequencer	Lane	Custom
Range	Minimum	Maximum	Steps	Shift
Map	Slot	Target	Depth	Custom
Root / Scale	Root	Scale	-	-

Speed: 1/4, 1/8, 1/16, 1/32 and 1/64 are available as straight, triplet (T) and dotted (D) values. T is two-thirds of the straight duration; D is one-and-a-half times it. First activation resets saved lane rates to 1/16. Later mode switches in the same session retain edited speeds.

Root and Scale: MDD uses the Song's shared root and scale. On the main grid, SCALE toggles quantization; SHIFT + SCALE enables scale mode and cycles Song scales. Inside a paged MDD menu, SCALE changes pages.

Root and scale edits can affect other scale-mode clips. A scale change may be rejected if existing Normal clip notes cannot fit it. Notes Random generates note numbers within the selected range; scale-mode notes are quantized during playback.

Gate controls note duration, not amp envelope decay or sustain.

Custom mapping

Choose a Custom lane, edit its steps, then open **Map > Slot > Target > Depth**. Each Custom has four target slots and starts with no assigned target.

Custom output is unipolar (0-100%). **+100% Depth** applies the full source amount; **+50%** applies half; **0%** disables its effect. Negative Depth reverses the direction.

The output modulates the base parameter setting; it does not assign an absolute target value. For LPF Frequency, positive Depth raises cutoff from its base value and negative Depth lowers it. Physical scaling and parameter limits still apply.

Target inactive means the target is not used by the current synth configuration. The popup lasts 0.8 seconds; selecting another inactive target restarts it. The list contains patchable parameters, not every synth or effect setting.

Plaits oscillator source

Select **Plaits** as the source for OSC 1 or OSC 2, then choose an engine. Existing oscillator sources remain available.

Engine	Character
1 - FM	Metallic, harmonic and percussive tones
2 - Waveshaping	Rich tones created through waveform shaping
3 - Virtual Analog	Analog-style oscillator sounds

Plaits control	Oscillator parameter field
Harmonics	Wave Index
Timbre	Pulse Width
Morph	Feedback

OSC 1 Engine / **OSC 2 Engine** in the Map list refers to Plaits engine selection, not Sine/Square/Saw selection. The corresponding oscillator must use Plaits for the target to affect its sound.

Custom lanes can modulate engine changes during a sequence. The resulting engine depends on the base Engine setting and Depth; a Custom step is not an absolute engine assignment. This integration contains three engines, not the complete Plaits feature set.

WRITE, Undo and Redo

WRITE > Replace stock notes? > Yes > press SELECT

WRITE transfers the Notes, Gates and Velocity result into the Normal grid and switches to Normal mode. It replaces the previous Normal notes. MDD lane data remains available. Custom modulation is not converted into parameter automation.

Action	Result
No / BACK	Cancel without replacing notes
Yes	Write the generated sequence and enter Normal mode
UNDO	Restore the previous Normal sequence
REDO	Restore the written sequence

Undo restores the previous complete note rows, including row expression/automation and sequence direction. Undo/Redo leaves the clip in Normal mode. If the required allocation fails, the write is aborted without replacing the existing notes.

Changes from the original firmware

Area	Addition or change
Sequencer	Clip-based MDD mode alongside the Normal sequencer
Modulation	Four Custom sources and mapping controls
Oscillators	Three-engine Plaits source for OSC 1 and OSC 2
Sequence transfer	WRITE with native Undo/Redo
OLED	MDD menus, step-edit notifications and selected long-text scrolling
Startup	Approximately four-second Holdontoentropy greeting after loading
Effects	Fixes to delay setting copies, granular processing and digital reverb behaviour

SHIFT + SELECT retains its existing Settings function. Reverb remains within the existing architecture. This release does not add a Valhalla/Eventide engine or independent insert reverb.

Current limitations

- MDD Kit sequencing is not supported.

- Custom MIDI CC/CV modulation output is not implemented.

- Long tie chains and overlapping-note sequencing are not implemented. Native portamento is used; strict legato requiring overlapping notes is not guaranteed.

- Plaits is limited to three engines.

- General OLED text layout improvements are unfinished. Scrolling currently covers specific WRITE and Map/Scale fields.

- Inactive targets can remain selectable.

Validation and attribution

The ARM build and software regression tests passed. Comprehensive physical Deluge testing has not been completed. Use backed-up project copies for initial testing.

Plaits uses a subset of Emilie Gillet's Mutable Instruments DSP source under the MIT license. Existing Deluge firmware license notices are retained.

Documentation describes the 10 October 2026 custom build, commit 8de154f7. Features described as limitations are not implemented in that build.

Bug fixes included in this build

The earlier general-firmware fixes are grouped into eight items below. These are separate from the MDD-specific integration fixes added during development. Grouping is for readability, rather than a claim that only eight lines or defects changed.

#	Fix	What changed
1	OLED submenu boundary	Walking backward through a submenu no longer constructs an iterator before the start of the item list. This removes an invalid boundary operation while drawing the previous menu entries.
2	Accelerated SELECT overflow	The temporary encoder offset was widened from 8-bit to 32-bit before applying SHIFT acceleration, preventing intermediate overflow and unintended navigation direction.
3	Delay sync setting copy	Copying Delay settings now also copies syncType, retaining the straight/triplet/dotted sync selection.
4	Granular buffer setup order	The grain buffer and effect state are prepared before shutdown-wrap timing is updated, avoiding use of unprepared timing state.
5	Granular grain-size calculation	Grain size is calculated using the newly updated grain rate rather than its previous value when the rate changes.
6	Saved reverb HPF restoration	Loading Song reverb settings now restores the stored high-pass filter setting alongside the other reverb parameters.
7	Reverb damping state separation	The second digital reverb branch uses its own low-pass damping state instead of reusing the first branch state.
8	Digital reverb width / initialization	Digital reverb now stores and applies stereo width; filter state and width have explicit initial values. Width zero folds wet output to mono.

Additional MDD fixes: preserve stock SHIFT + SELECT routing; retain synth-menu access; protect active Custom values when another clip sharing the synth closes; preserve pending mode state when cloning; handle sample-counter rollover in long-press timing; clear stale grid lights; correct initial 1/16 timing, scale handling, Depth percentage rounding and WRITE Undo/Redo.

Verification is based on the delivered source diff and regression tests. These fixes do not establish that every firmware issue has been eliminated. Physical-device validation remains outstanding.

Where Plaits comes from

Plaits is an open-source synthesizer originally developed by **Emilie Gillet / Mutable Instruments**. This firmware reuses its DSP code instead of implementing an imitation from scratch. Only a selected subset of the source is included.

Source repositories: github.com/pichenettes/eurorack and the supporting github.com/pichenettes/stmlib library.

Dependency	Pinned source revision
eurorack / Plaits	08460a69a7e1f7a81c5a2abcc7189c9a6b7208d4
stmlib	e3bd7c9cc00e4364166f9905c0509b6ffd0535ec

The vendored subset contains the **FM, Waveshaping and Virtual Analog** engines and their required supporting code. Upstream engine files are retained unchanged, with their MIT license notices. Deluge-specific integration lives in a separate adapter and oscillator/voice code.

How it was integrated

Oscillator source: Plaits is added alongside existing Deluge sources for OSC 1 and OSC 2. It runs inside the Deluge voice path, using Deluge envelopes, filters and effects; this is not a complete standalone Plaits voice recreation.

Pitch and controls: the adapter converts Deluge phase increments to Plaits pitch, compensating for the Plaits reference sample rate versus the Deluge 44.1 kHz output. Harmonics, Timbre and Morph receive normalized values from the corresponding oscillator controls.

Engine parameter: a patchable Engine parameter selects the three engines, displayed as 1-3. This connects engine changes to the native modulation system and MDD Custom sources.

Live switching: engine instances reuse reserved storage. Switching engines does not allocate new heap memory. A 64-sample transition from the preceding output value softens the change; it is not a full crossfade between two simultaneously running engines.

Saving: oscillator source and Engine settings are integrated into the firmware parameter and serialization paths so the selected sound can be stored.

Scope and licensing

The full Plaits engine collection, original hardware interface and complete trigger/LPG behaviour are outside this integration. The source code is MIT-licensed, but that does not imply Mutable Instruments endorsement of this custom firmware. Existing Deluge license notices remain applicable.